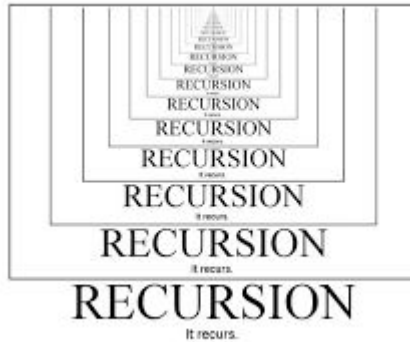


Fundamentals of Recursion



Lecture Flow

1. Pre-requisites
2. Problem definitions
3. Basic Concepts
4. Time and Space Complexity of Data Structure
5. Common pitfalls
6. Practice questions and Resources
7. Quote of the day

Pre-requisites

- Understanding iterative program flow
- Functions, local variables and global variables
- Stack data structures
- Willingness to learn

Thinking recursively

Recursion is an important concept in computer science. It is a foundation for many other algorithms and data structures. However, the concept of recursion can be tricky to grasp for many beginners.

Real Life Example

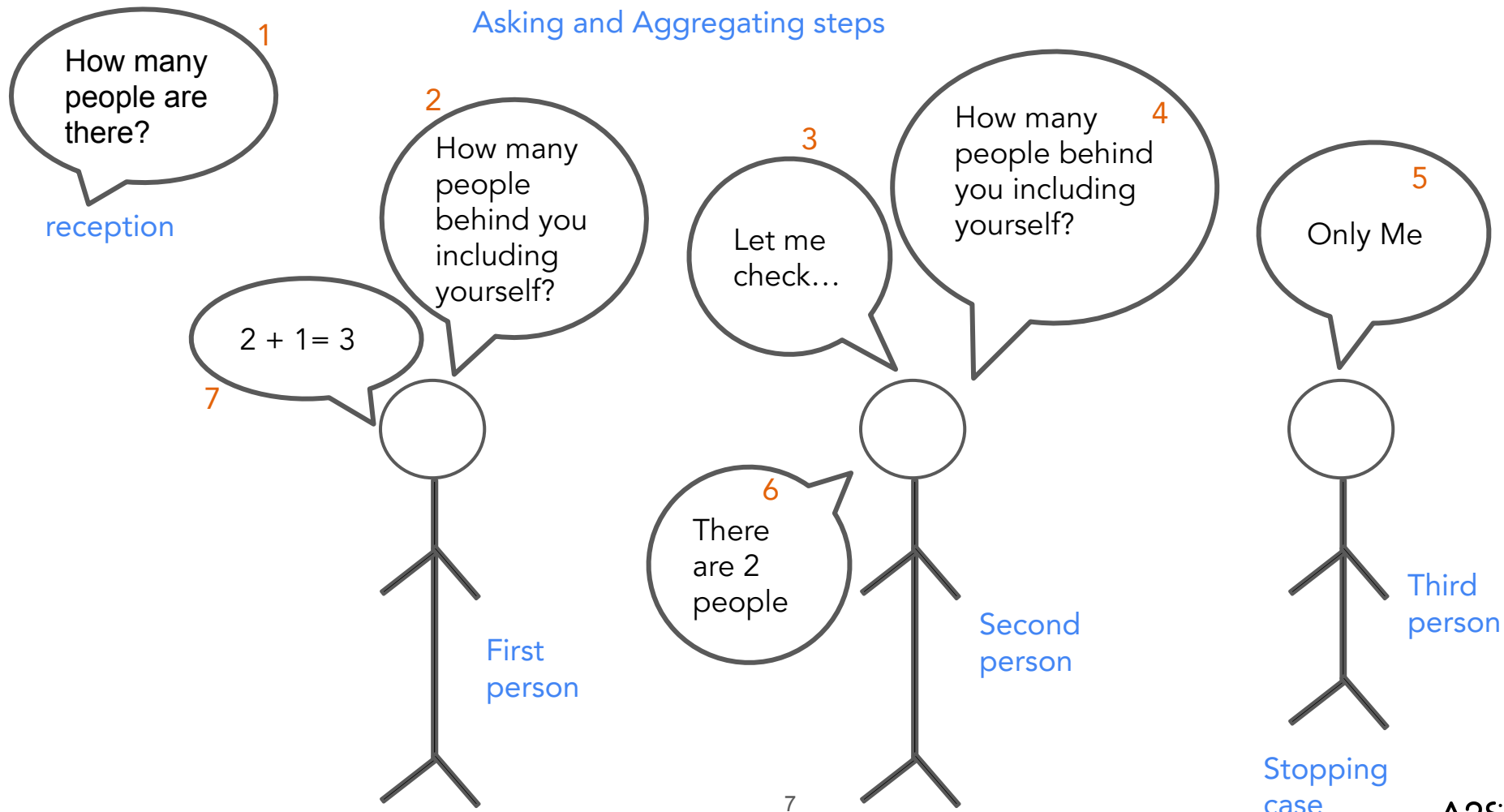
Imagine you are in a long queue and the reception asks you how many people there are in the line. How do you count the number of people in the queue?



Brainstorm

- Let's assume each person in the queue
 - Can increment numbers by 1
 - Can request other people
- What if I **asked the person behind** me "how many people there are" and whatever their response is, I can **do plus 1** and inform the reception? And
- What if the **person I asked, asks the person behind them** "how many people there are" and whatever they get, they do **plus 1** and **inform me**?
 -
 -
 -

Asking and Aggregating steps



Can we simulate this with code?

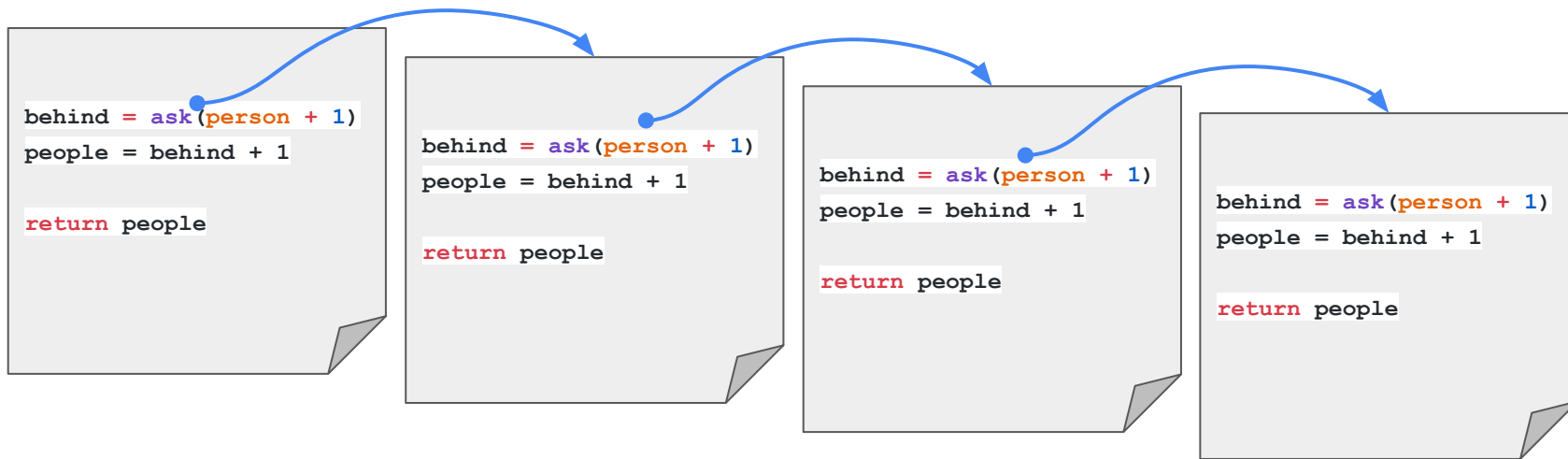
- Since we are doing the same thing again and again let's use **one function** to do the task



Does this work?

```
def ask(person):  
  
    behind = ask(person + 1)  
    people = behind + 1  
  
    return people
```

What happens?

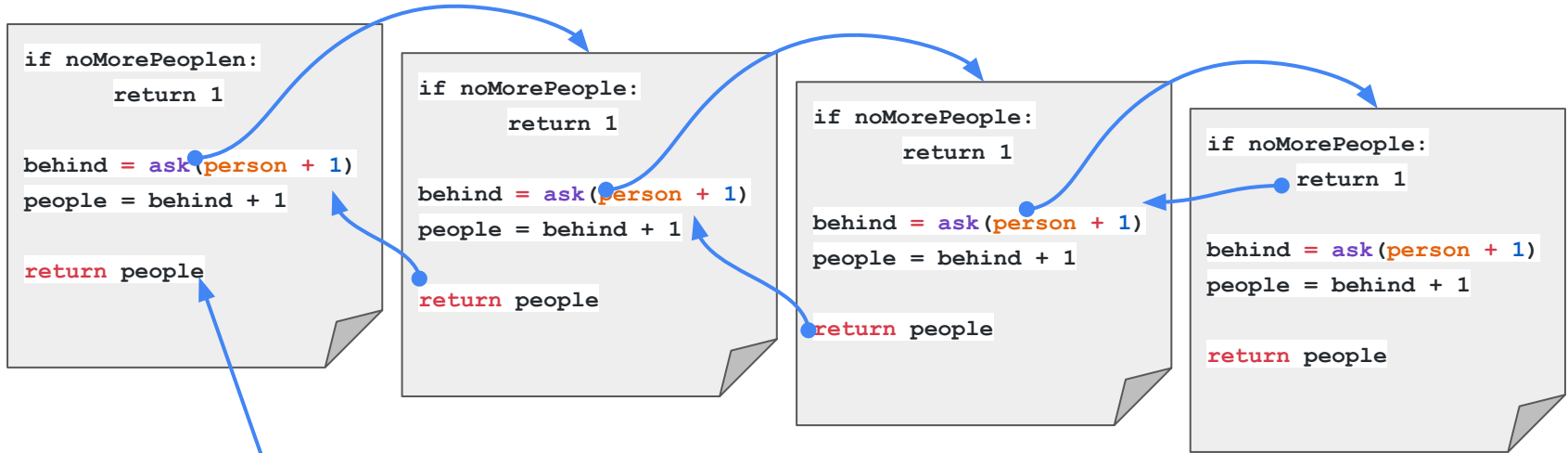


We will **never get to return** because we are calling the function non stop. What should we do?

Does this work?

```
def ask(person):  
    if noMorePeople:  
        return 1  
  
    behind = ask(person + 1)  
    people = behind + 1  
  
    return people
```

What happens?



Then now we return the number of people to our first caller

Real Life Example 2

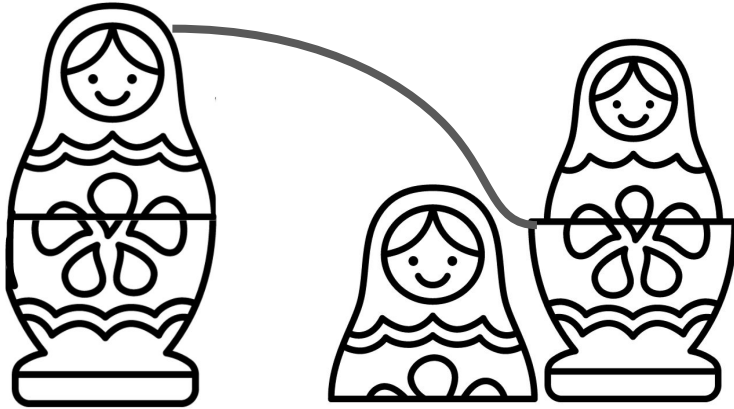
Imagine some one give you a matryoshka and ask you to count the number dolls, how would you solve the problem?



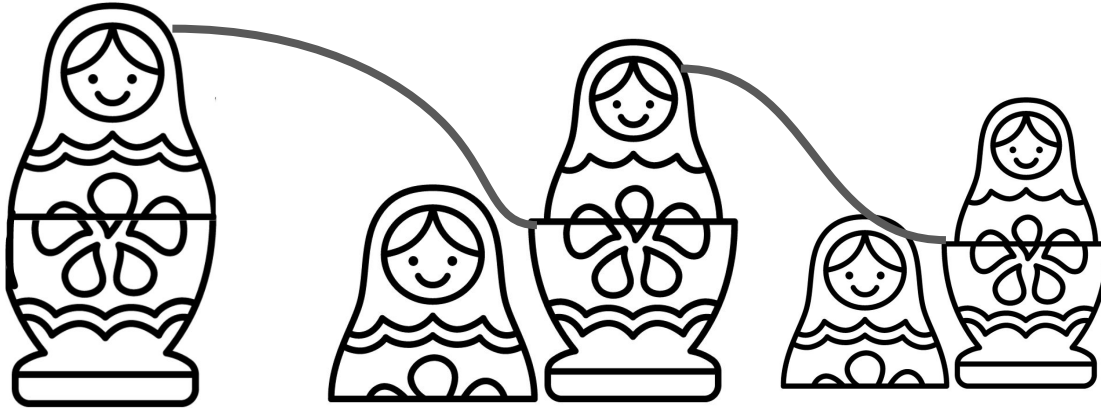
Brainstorm



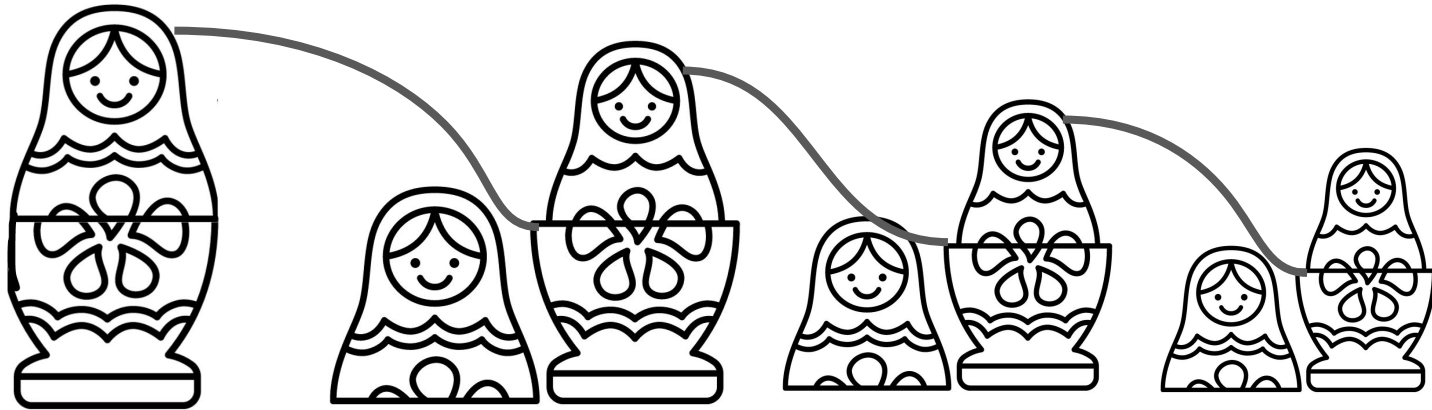
Brainstorm



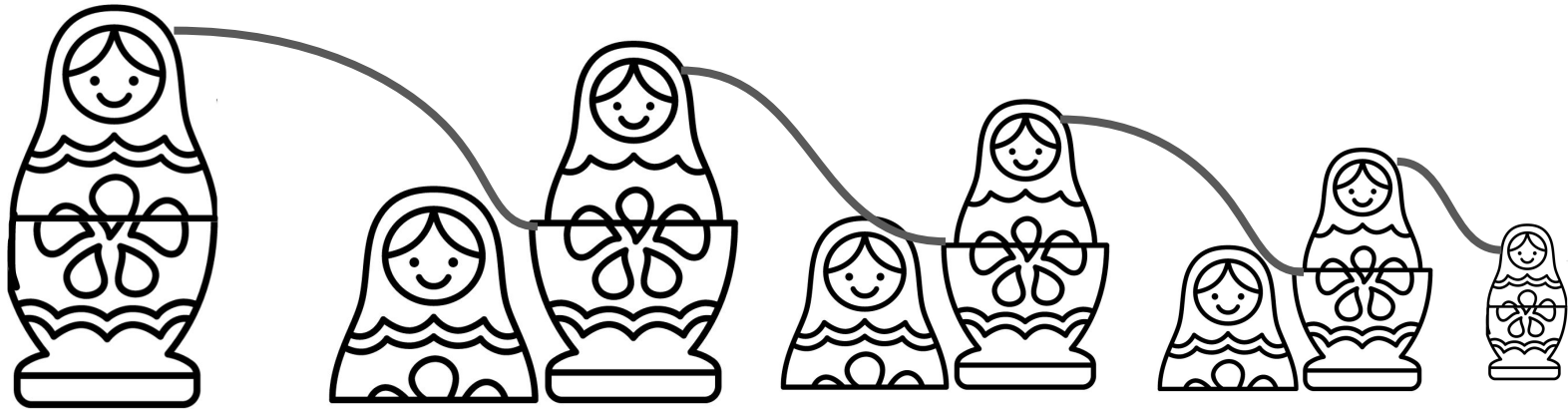
Brainstorm



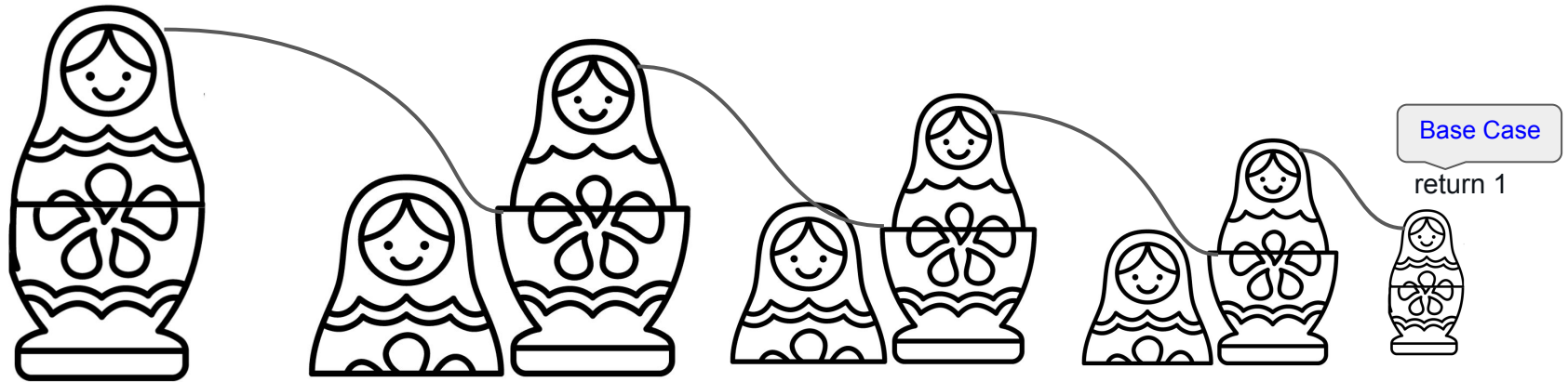
Brainstorm



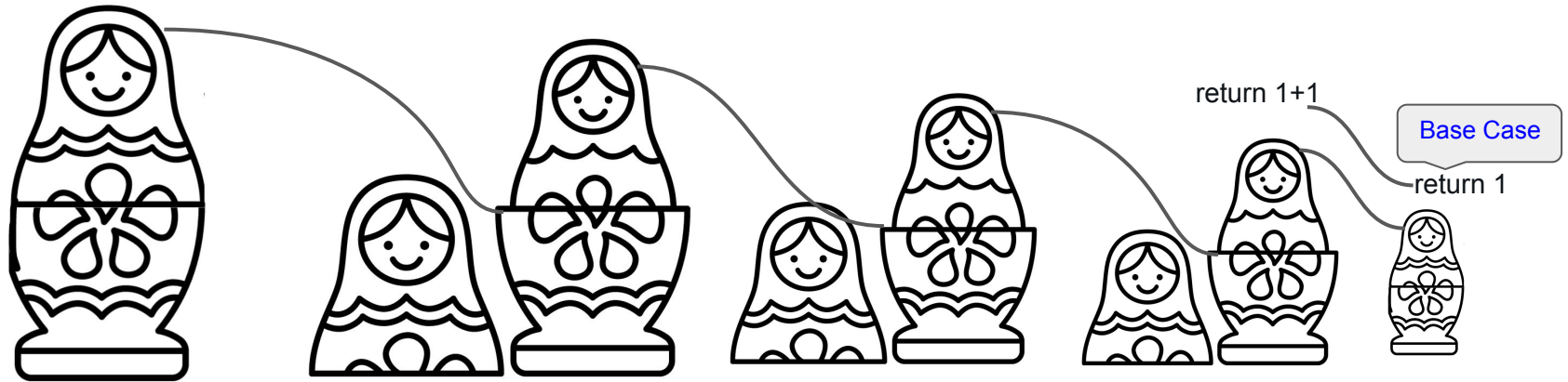
Brainstorm



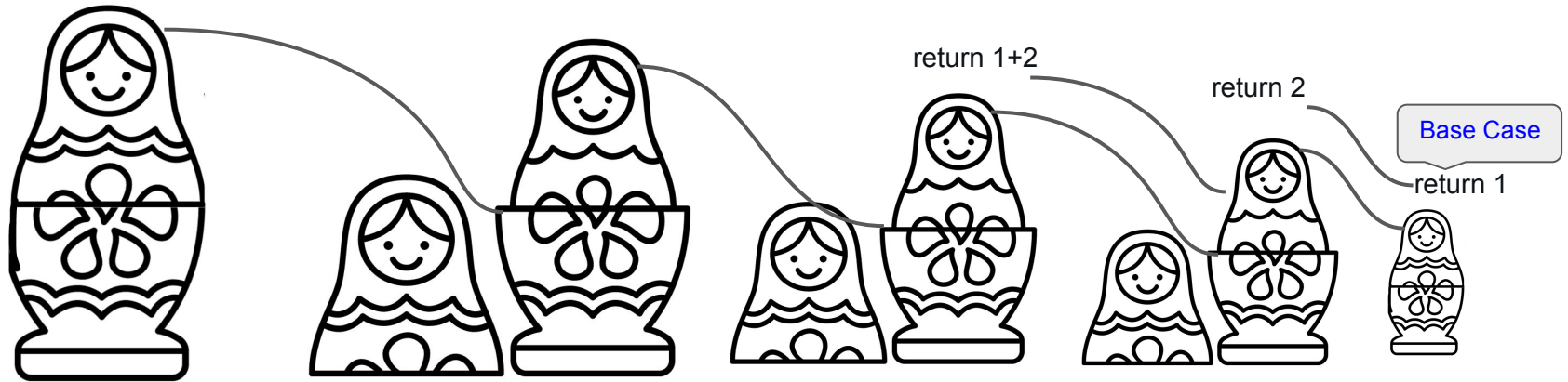
Brainstorm



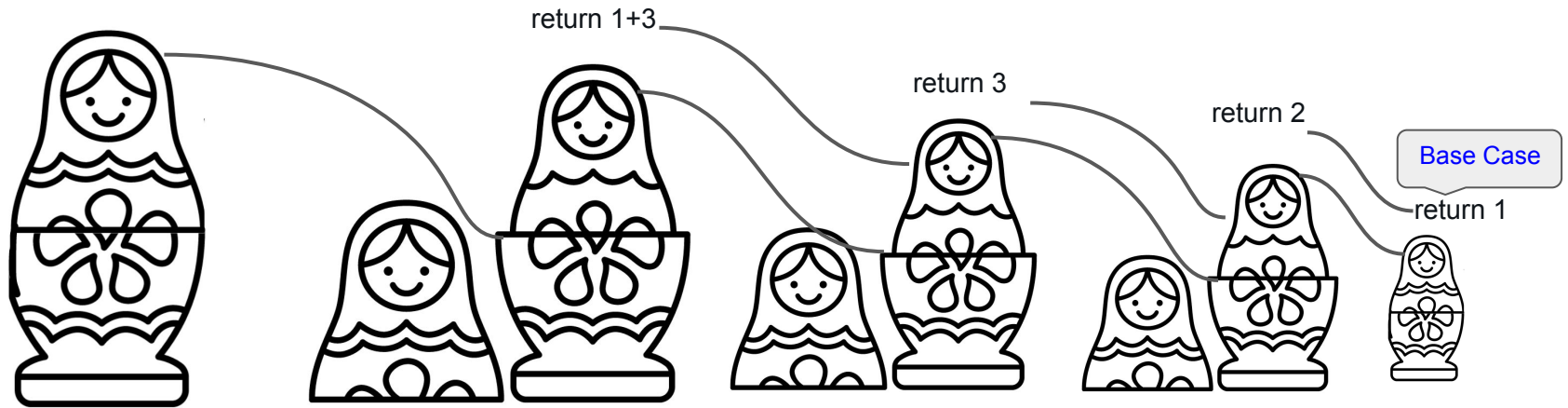
Brainstorm



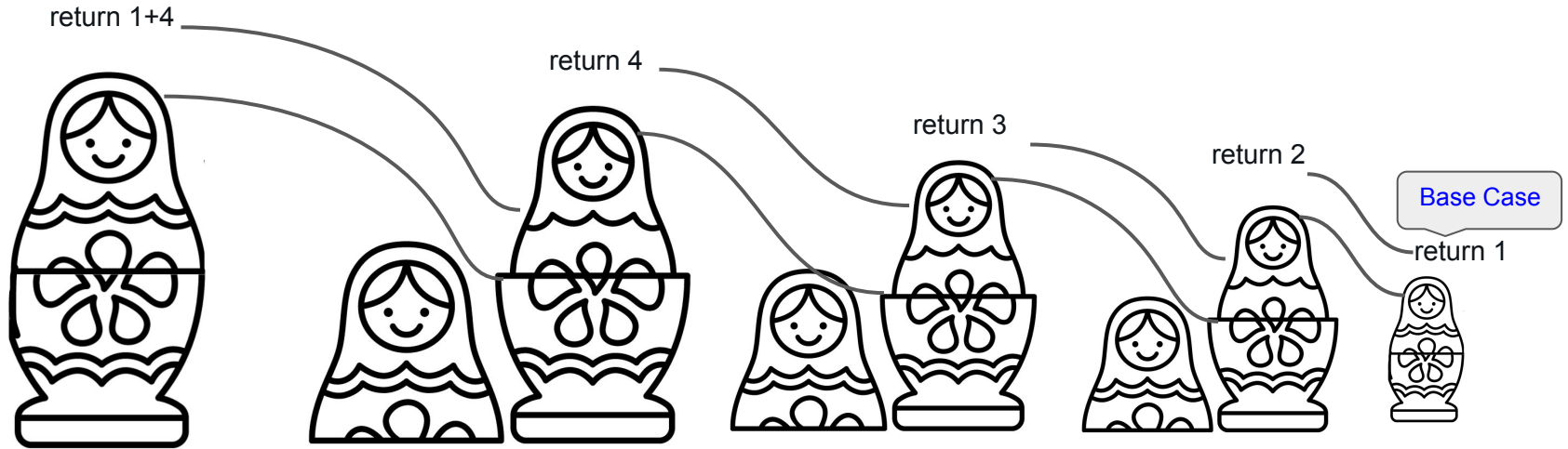
Brainstorm



Brainstorm



Brainstorm



Can we simulate this with code?

- Since we are doing the same thing again and again let's use **one function** to do the task

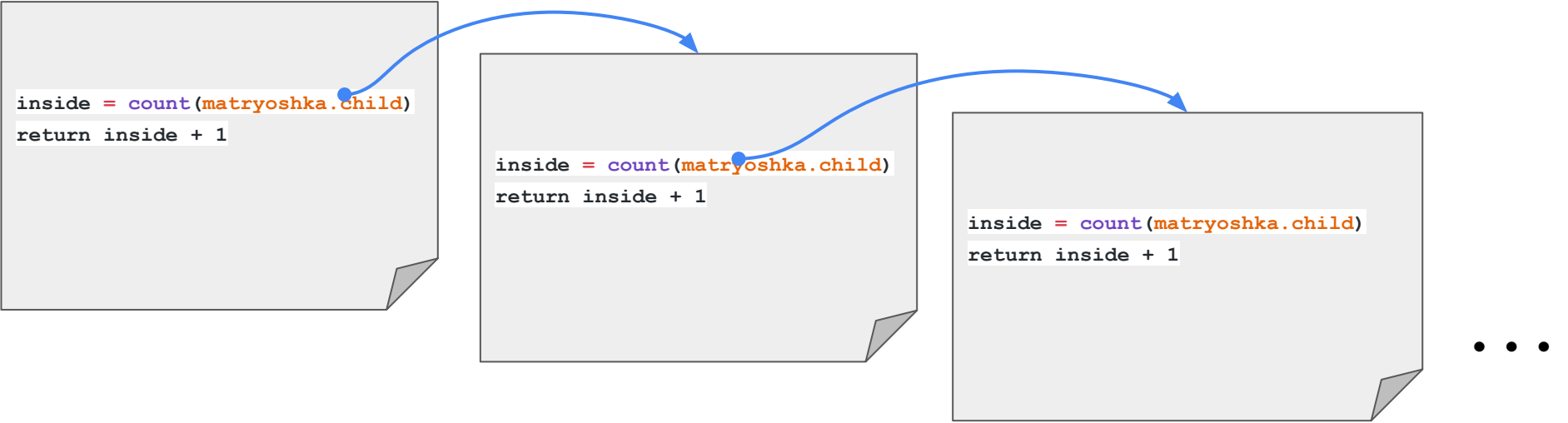


Does this work?

```
def count(matryoshka):  
  
    inside = count(matryoshka.child)  
  
    return inside + 1
```

What happens?

```
inside = count(matryoshka.child)  
return inside + 1
```



```
inside = count(matryoshka.child)  
return inside + 1
```

```
inside = count(matryoshka.child)  
return inside + 1
```

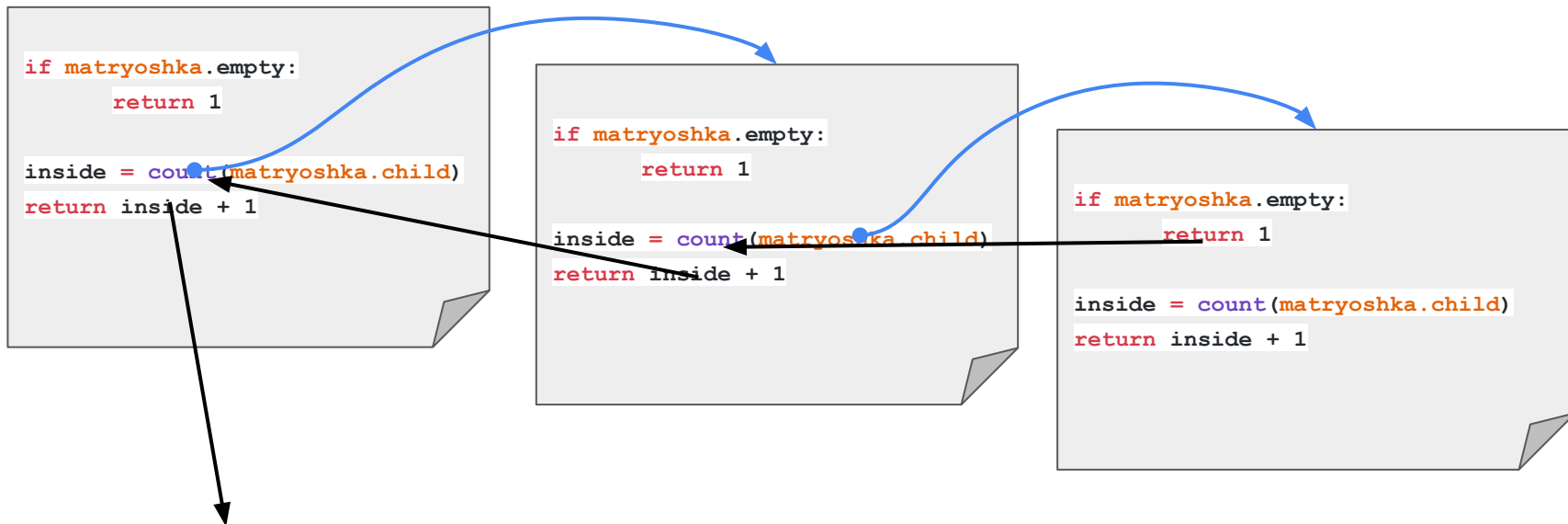
...

We will **never get to return** because we are calling the function non stop. What should we do?

How about this?

```
def count(matryoshka):  
    if matryoshka.empty:  
        return 1  
  
    inside = count(matryoshka.child)  
  
    return inside + 1
```

What happens?



Then now we return the number of people to our first caller

Definitions



Recursion: process in which a function calls itself directly or indirectly

Basic Concept

The basic concept of recursion is that a problem can be solved intuitively and much easily if it is represented in one or more smaller versions.

What was the problem in the previous real life example?

What was the subproblem?



Basic components of recursion

- **Base case** The condition that signals when the function should stop and return the final state.
- **Recurrence relation** Reduces all cases towards base case. The section where the function calls itself with modified inputs and state
- **State** An identifier that fully locate which subproblem we are dealing with currently

What were the **Base Case**, **Recursive Relation** and **The State** in the previous example?

```
def count(matryoshka):  
    if matryoshka.empty:  
        return 1  
  
    inside = count(matryoshka.child)  
  
    return inside + 1
```

State

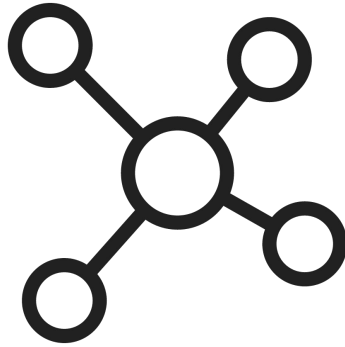
Base Case

Recurrence Relation



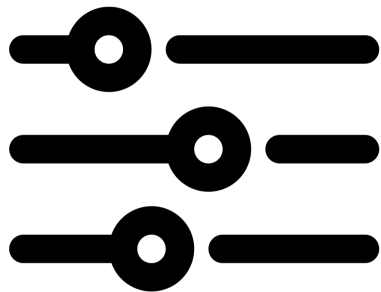
Base case

- are **known** and **simplest** cases of the problem
- is a **terminating** scenario
- initiate the process of returning to the original call function (or problem)



Recurrence Relation

- should be a **breakdown** of the current problem into smaller problems
- Is a set of cases **reduces the current case** towards base case
- It **aggregates** the result from recursive calls and **returns** the answer we have so far **to caller**



State

- Identifies the current subproblem completely
- Helps locate which subproblem we are dealing with currently
- Changes in state should lead towards base cases

Sample Question

509. Fibonacci Number

Solved 

Easy



Topics



Companies

The **Fibonacci numbers**, commonly denoted $F(n)$ form a sequence, called the **Fibonacci sequence**, such that each number is the sum of the two preceding ones, starting from 0 and 1 . That is,

$$F(0) = 0, F(1) = 1$$

$$F(n) = F(n - 1) + F(n - 2), \text{ for } n > 1.$$

Given n , calculate $F(n)$.

Base Case

- If $n == 1$, return 1
- If $n == 0$, return 0

State

- n

Recurrence Relation

- $f(n) = f(n - 1) + f(n - 2)$

Practice

Power of Four

What is the base case and recurrence relation for the above problem?

Solution

Q1. What happens if we **continue to divide** a number which is a power of four **by four**?

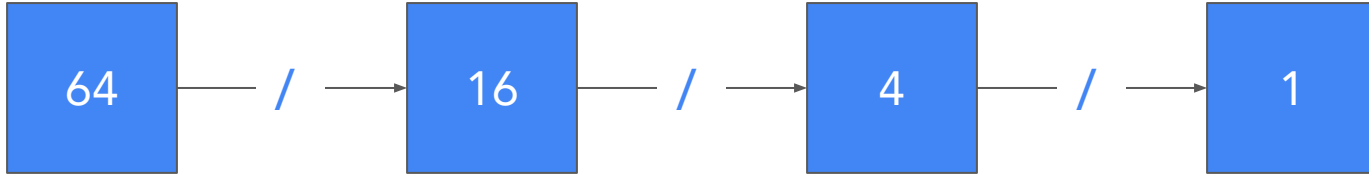
- The final result will be 1

Q2. What if the number **is not power of four**?

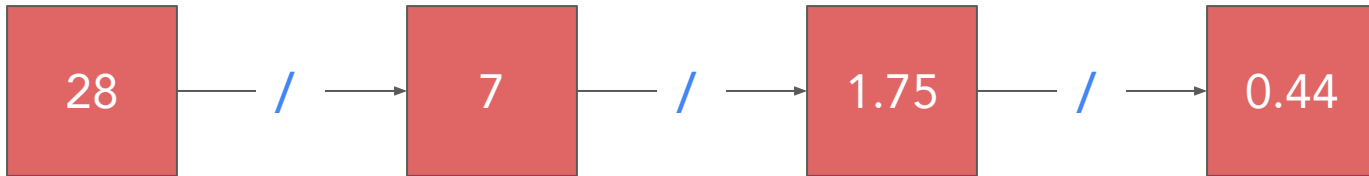
- Their number will be below 1 eventually

From the above questions we can see that we have **two base cases** and **the state is the number** that is divided by four every time we have recursive call.

Dividing a power of four by four



Dividing a random number by four





How does recursion work?

How does the recursion work?

- The system uses some ordering to execute the calls
- The ordering is done based on Last In First Out (depth first) order
- It uses stack

`Stack.top` -> currently executing

`Stack.push` -> calling function

`Stack.pop` -> returning a value

Example Using Call Stack

```
def fib(n: int) -> int:
    if n == 1:
        return 1

    if n == 0:
        return 0

    one = fib(n-1)
    two = fib(n-2)

    return one + two
```

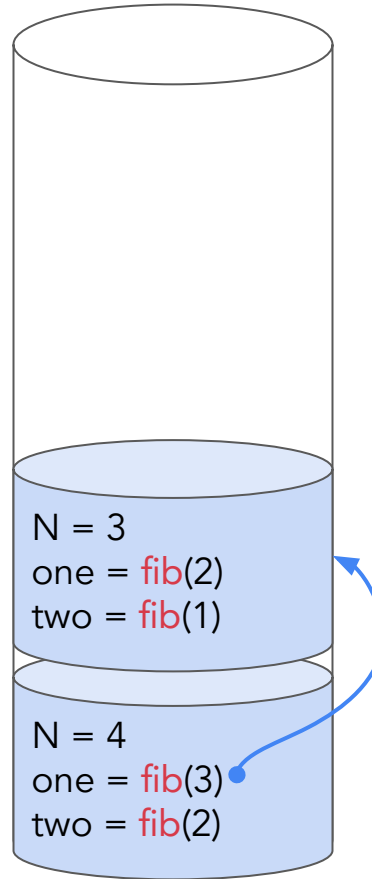


```
def fib(n: int) -> int:
    if n == 1:
        return 1

    if n == 0:
        return 0

    one = fib(n-1)
    two = fib(n-2)

    return one + two
```

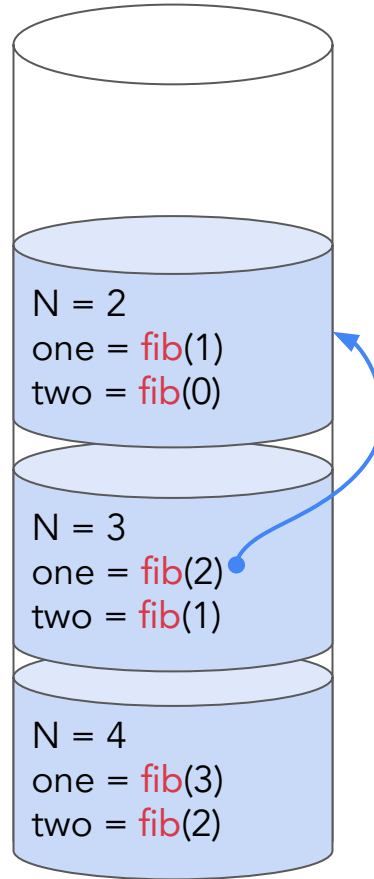


```
def fib(n: int) -> int:
    if n == 1:
        return 1

    if n == 0:
        return 0

    one = fib(n-1)
    two = fib(n-2)

    return one + two
```

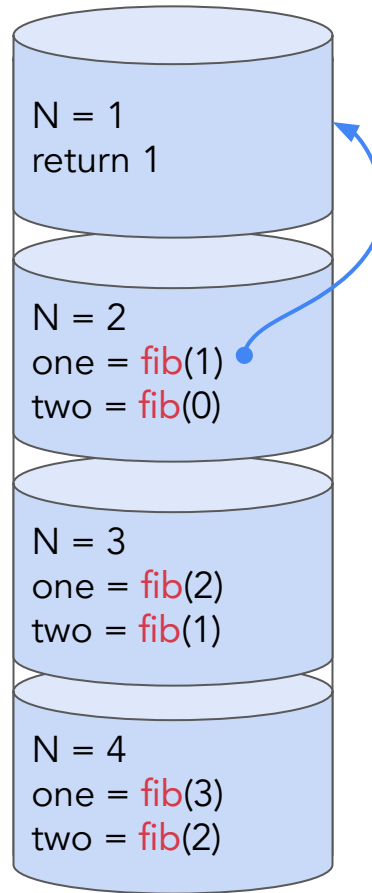


```
def fib(n: int) -> int:
    if n == 1:
        return 1

    if n == 0:
        return 0

    one = fib(n-1)
    two = fib(n-2)

    return one + two
```

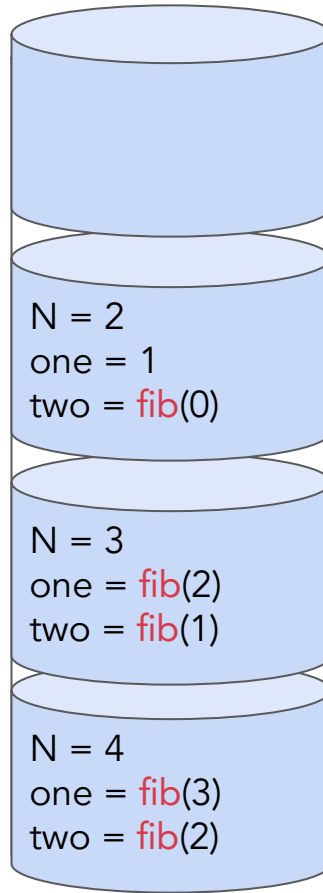


```
def fib(n: int) -> int:
    if n == 1:
        return 1

    if n == 0:
        return 0

    one = fib(n-1)
    two = fib(n-2)

    return one + two
```

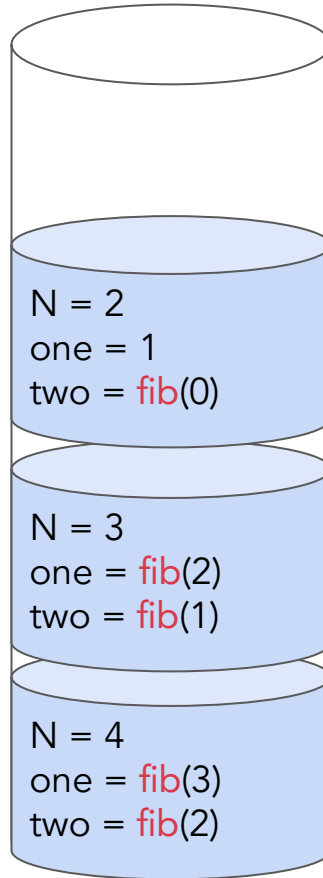


```
def fib(n: int) -> int:
    if n == 1:
        return 1

    if n == 0:
        return 0

    one = fib(n-1)
    two = fib(n-2)

    return one + two
```

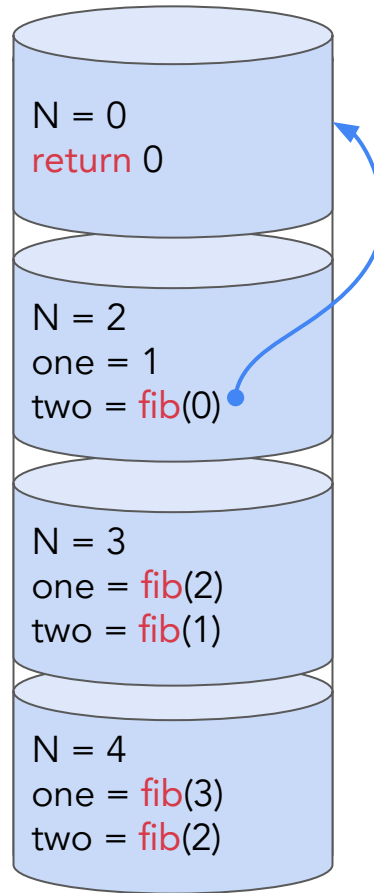


```
def fib(n: int) -> int:
    if n == 1:
        return 1

    if n == 0:
        return 0

    one = fib(n-1)
    two = fib(n-2)

    return one + two
```

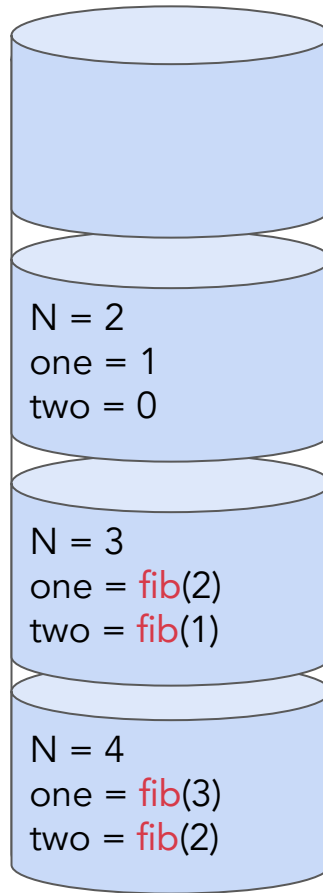


```
def fib(n: int) -> int:
    if n == 1:
        return 1

    if n == 0:
        return 0

    one = fib(n-1)
    two = fib(n-2)

    return one + two
```

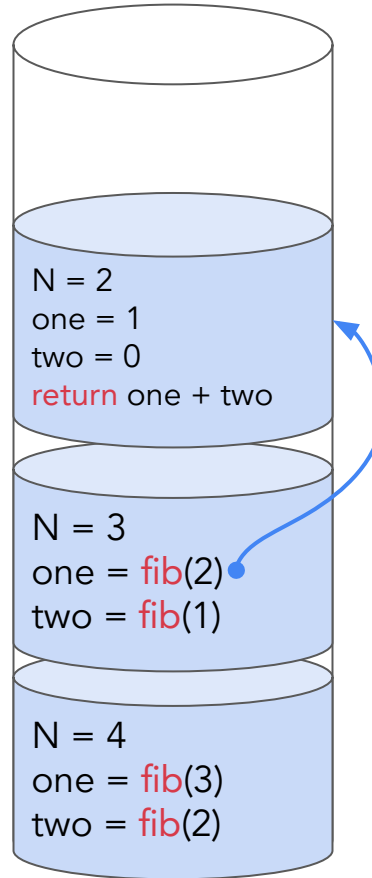


```
def fib(n: int) -> int:
    if n == 1:
        return 1

    if n == 0:
        return 0

    one = fib(n-1)
    two = fib(n-2)

    return one + two
```

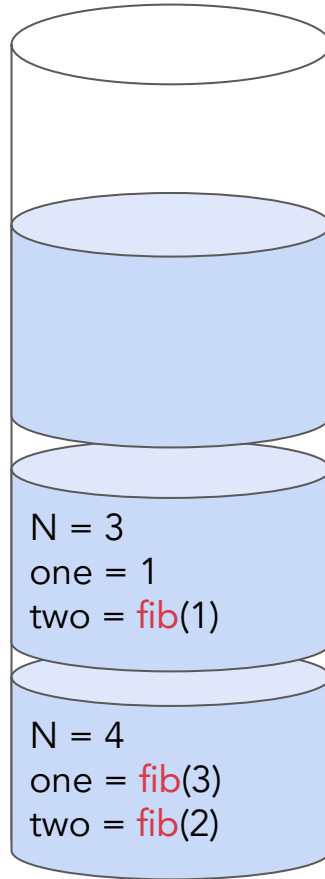


```
def fib(n: int) -> int:
    if n == 1:
        return 1

    if n == 0:
        return 0

    one = fib(n-1)
    two = fib(n-2)

    return one + two
```

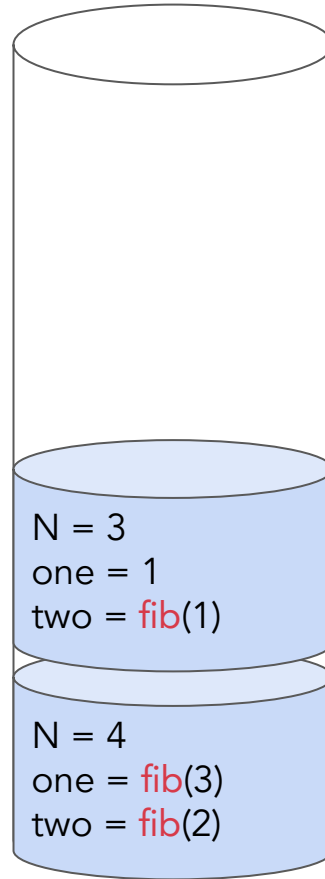


```
def fib(n: int) -> int:
    if n == 1:
        return 1

    if n == 0:
        return 0

    one = fib(n-1)
    two = fib(n-2)

    return one + two
```

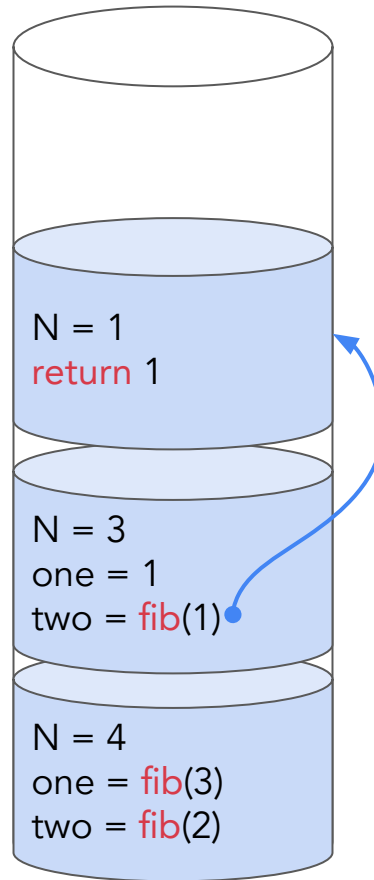


```
def fib(n: int) -> int:
    if n == 1:
        return 1

    if n == 0:
        return 0

    one = fib(n-1)
    two = fib(n-2)

    return one + two
```

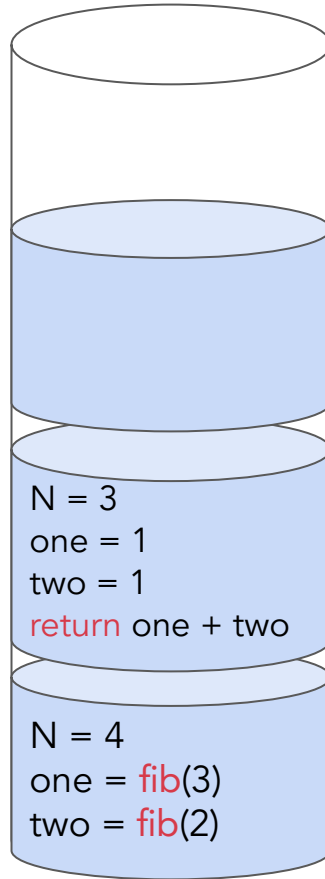


```
def fib(n: int) -> int:
    if n == 1:
        return 1

    if n == 0:
        return 0

    one = fib(n-1)
    two = fib(n-2)

    return one + two
```

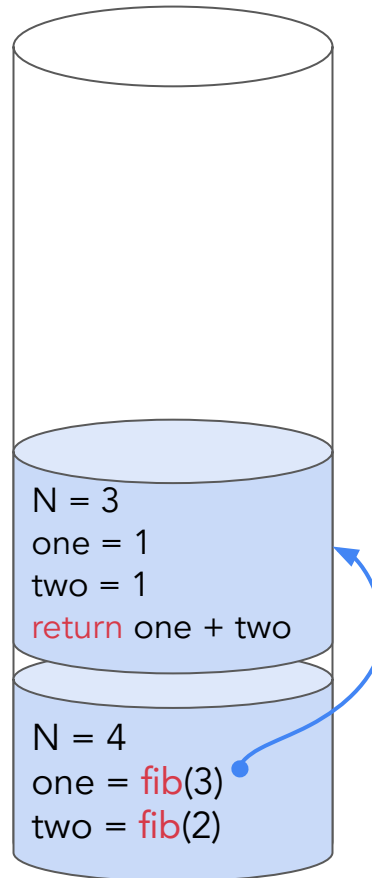


```
def fib(n: int) -> int:
    if n == 1:
        return 1

    if n == 0:
        return 0

    one = fib(n-1)
    two = fib(n-2)

    return one + two
```

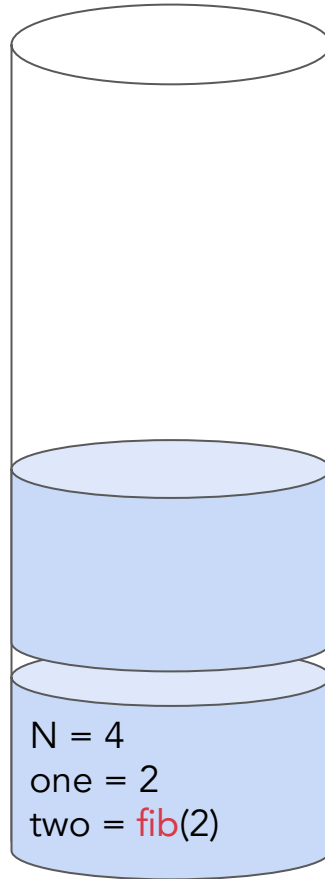


```
def fib(n: int) -> int:
    if n == 1:
        return 1

    if n == 0:
        return 0

    one = fib(n-1)
    two = fib(n-2)

    return one + two
```



```
def fib(n: int) -> int:
    if n == 1:
        return 1

    if n == 0:
        return 0

    one = fib(n-1)
    two = fib(n-2)

    return one + two
```

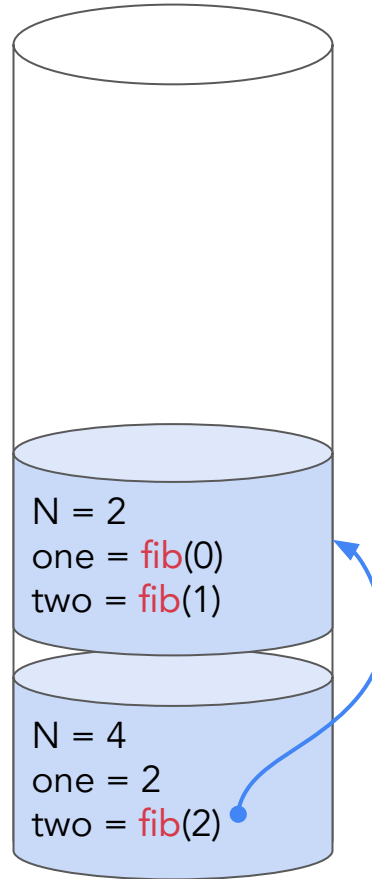


```
def fib(n: int) -> int:
    if n == 1:
        return 1

    if n == 0:
        return 0

    one = fib(n-1)
    two = fib(n-2)

    return one + two
```

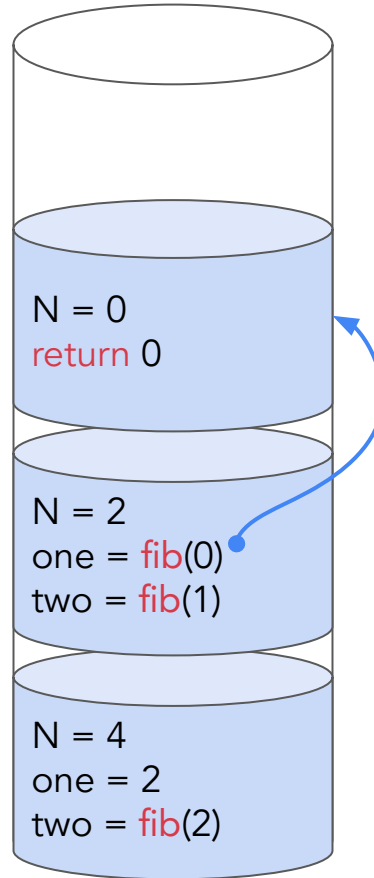


```
def fib(n: int) -> int:
    if n == 1:
        return 1

    if n == 0:
        return 0

    one = fib(n-1)
    two = fib(n-2)

    return one + two
```

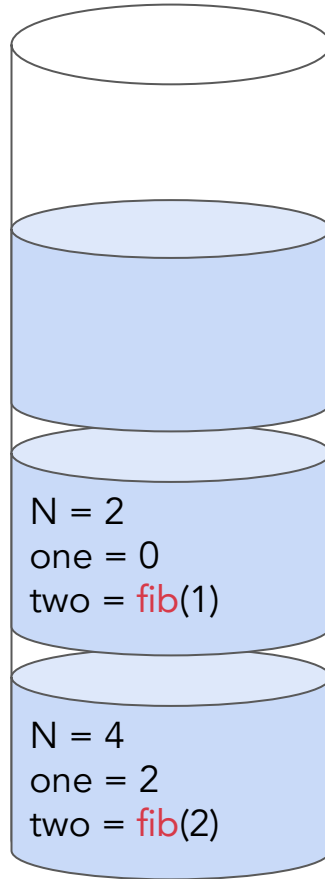


```
def fib(n: int) -> int:
    if n == 1:
        return 1

    if n == 0:
        return 0

    one = fib(n-1)
    two = fib(n-2)

    return one + two
```

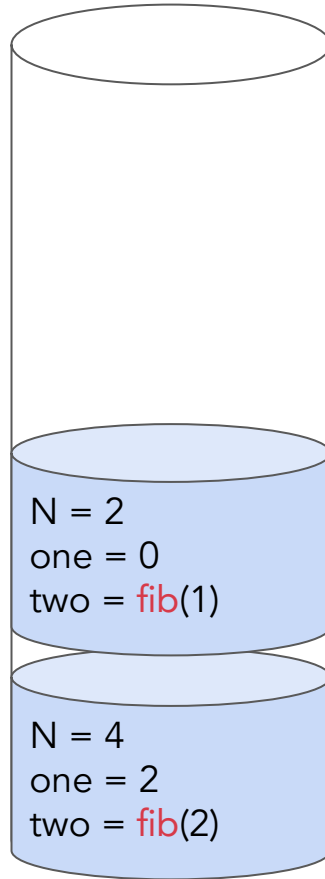


```
def fib(n: int) -> int:
    if n == 1:
        return 1

    if n == 0:
        return 0

    one = fib(n-1)
    two = fib(n-2)

    return one + two
```

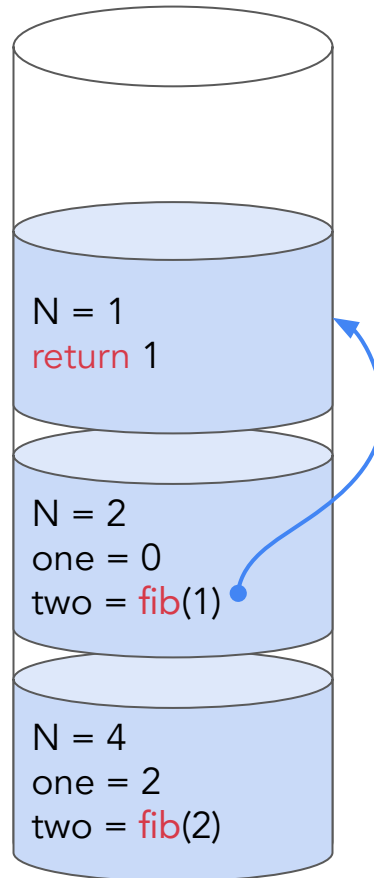


```
def fib(n: int) -> int:
    if n == 1:
        return 1

    if n == 0:
        return 0

    one = fib(n-1)
    two = fib(n-2)

    return one + two
```

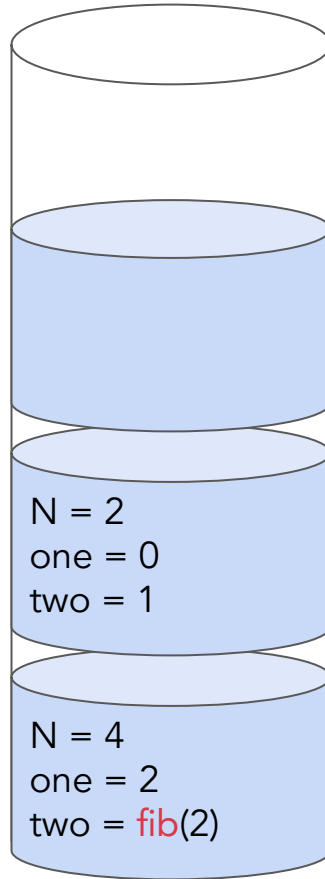


```
def fib(n: int) -> int:
    if n == 1:
        return 1

    if n == 0:
        return 0

    one = fib(n-1)
    two = fib(n-2)

    return one + two
```

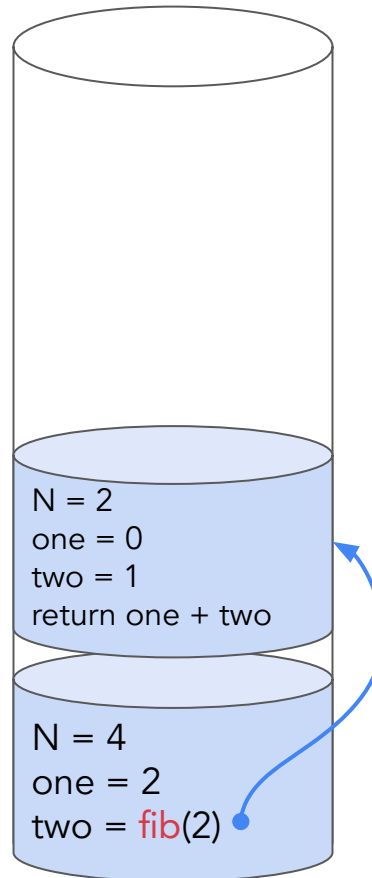


```
def fib(n: int) -> int:
    if n == 1:
        return 1

    if n == 0:
        return 0

    one = fib(n-1)
    two = fib(n-2)

    return one + two
```

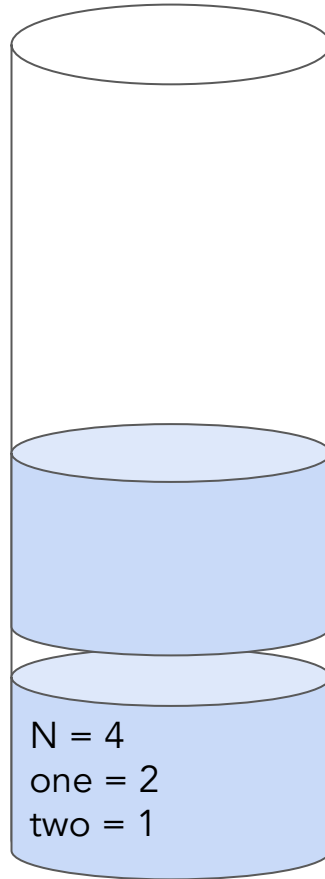


```
def fib(n: int) -> int:
    if n == 1:
        return 1

    if n == 0:
        return 0

    one = fib(n-1)
    two = fib(n-2)

    return one + two
```



```
def fib(n: int) -> int:
    if n == 1:
        return 1

    if n == 0:
        return 0

    one = fib(n-1)
    two = fib(n-2)

    return one + two
```



Using Execution Tree

Step by Step Simulation

Go to this [LINK](#)

The screenshot shows the 'Step by Step Simulation' web application. The interface is divided into a left sidebar and a main workspace. The sidebar contains several sections: a header with a logo and text, a 'Pre-defined templates' dropdown menu (highlighted with a red box), 'Global variables' input fields, a 'Recursive function' code editor (with a 'python' language selector), 'Options' with toggle switches for 'Step-by-step animation', 'Memoization', and 'Dark mode', and a bottom input field for the function call 'fn(5)' (highlighted with a red box) next to a 'Run' button. The main workspace features a large diagram of three interconnected circles, navigation arrows at the top (the left arrow is highlighted with a red box), and a 'Show desktop' button at the bottom right. Red arrows point from the highlighted elements in the sidebar to the main workspace area.

Your new development career awaits.
Check out the latest listings.

ADS VIA CARBON

Pre-defined templates

Fibonacci

Global variables

Recursive function python

```
def fn(n):  
    if (n == 0 or n == 1):  
        return n  
  
    return fn(n-1) + fn(n-2)
```

Options

Step-by-step animation ☒

Memoization ☐

Dark mode ☐

fn(5) **Run**

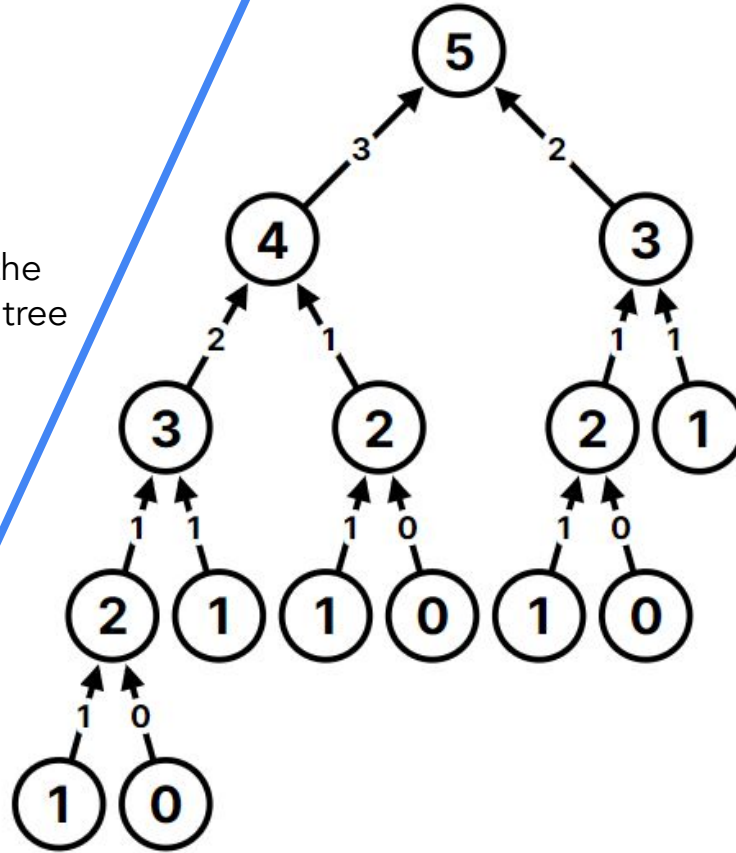
« < > »

Made with ♥ by Bruno Papa • [Github](#)

Show desktop

fn(5) returns 5

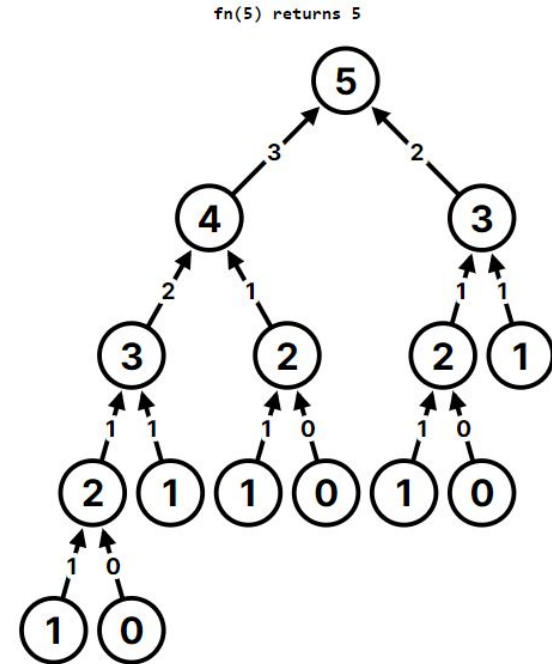
Depth of the
execution tree



Num of nodes
approximately
: $2^{n+1} - 1$

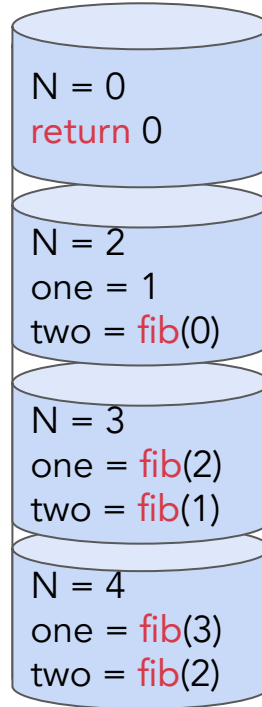
Debugging using the execution tree

- Simulate the code using the execution tree using pen and paper by yourself.
- Simulate your code with recursion simulation sites
- Compare the results



Debugging using the call stack and print statements

- Simulate the code using the call stack by yourself using pen and paper
- Use print statements to see if your code matches the simulation



```
def recursive_fn(state):  
    print("state: ", state)  
  
    .  
    .  
    .  
  
    print("result: ", result)  
    return result
```

Complexity Analysis



Time Complexity
(Any suggestion?)

There are two parts

- Number of recursive calls
- The complexity of the function

How do we find the number of calls?

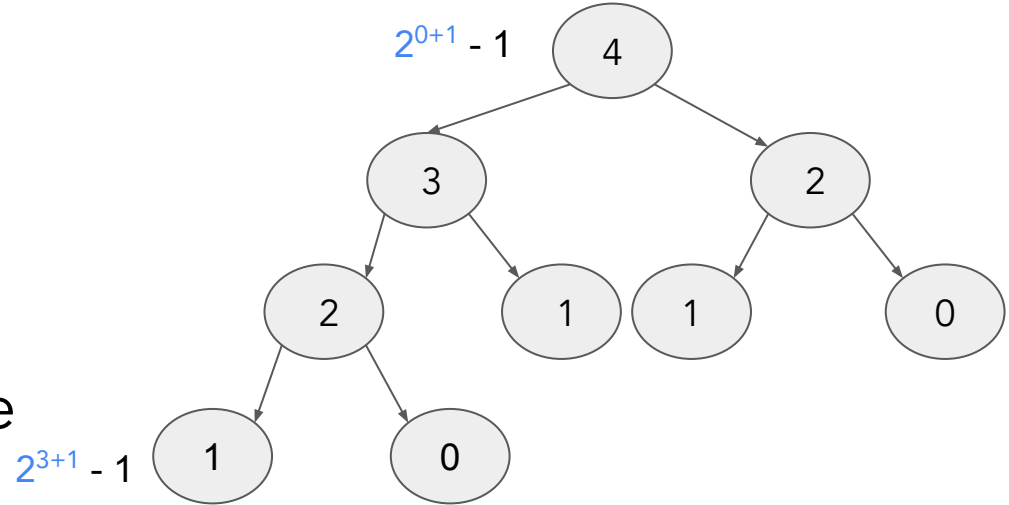
- Draw the **execution tree**
- Execution tree is a tree that describes the **flow of execution** for the recursive function
- A node represents a call
- The **total number of nodes** is your final answer



Fib(4)

Facts:

- Two branching on each node
- Height of the tree is the equal to n
- In a complete binary tree with n height, we have $2^{n+1} - 1$ nodes



In general

- Let
 - b = number of branches on each node
 - d = depth of the tree
- Total number of nodes = $b^{d+1} - 1$
- After only taking the dominant term
(Num of branching)^{depth of the tree}

What is the complexity of the function?

- This is plain time and space complexity analysis of the inner working of the function
- Are we doing any costly operations in the function?



In Summary

$$\underline{O(\text{function})} \bullet \underline{O((\text{Num of branching})^{\text{depth of the tree}})}$$

Cost of running function

Number of recursive calls

Space Complexity

Before we discuss the space complexity, let's consider these questions

- How does the computer run recursion function?
- How does it know when to return and combine?



It uses call stack



What is the space complexity of the **call stack**?



Maximum size of the stack which is maximum depth of the execution tree



Did we forget anything?

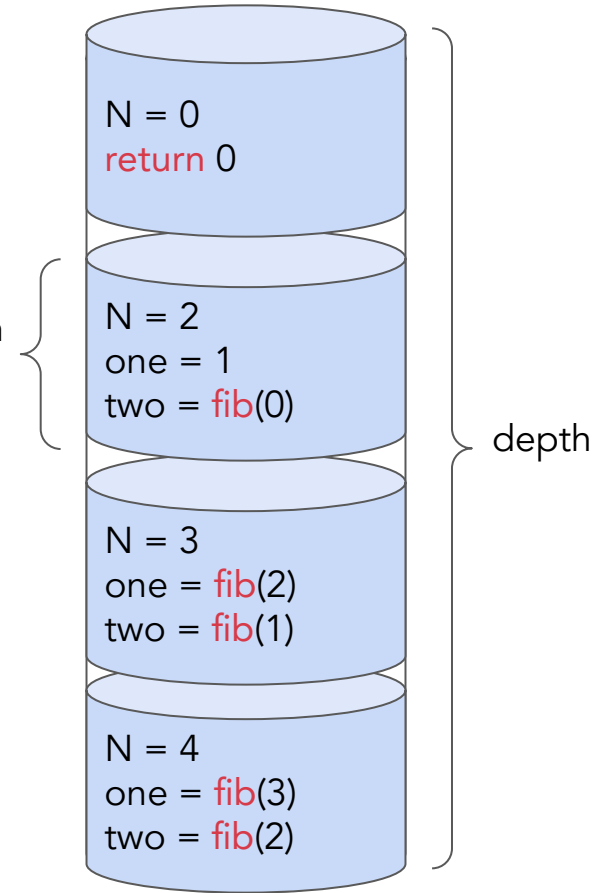


What about the **states** and **space complexity**
of the function?

In Summary

$[O(\text{states' size}) + O(\text{space complexity of function})] \bullet$
 $O(\text{maximum depth of the execution tree})$

State Size + function
complexity



Checklist (Ask your instructor, if anything is missing)

- ☐ You have to be very comfortable with **writing out the base case and recurrence relation** of recursive function
- ☐ You have to be very comfortable with **drawing execution tree** of a recursion function
- ☐ **Understanding the execution order** using the execution tree
- ☐ Comfortably **analyze the time and space complexity** of a recursive function

Recursion versus Iteration

Anything that can be done recursively can be done iteratively

- Iteration is faster and more space-efficient than recursion.
- Recursion has function call overhead
- Iterative programs are easy to debug

So why do we even need recursion?

- If the solution to the given problem is achieved by breaking it down into its subproblems, the recursive approach is a more intuitive and natural way to solve it.

Recognizing in Questions

- Look for patterns in the question that suggest that the **same operation is being applied to smaller versions** of the same problem.
- Look for **base cases or stopping conditions** that define when the recursion should end.
- **Searching problems with branching conditions**



Things to pay attention

Stack Overflow ?

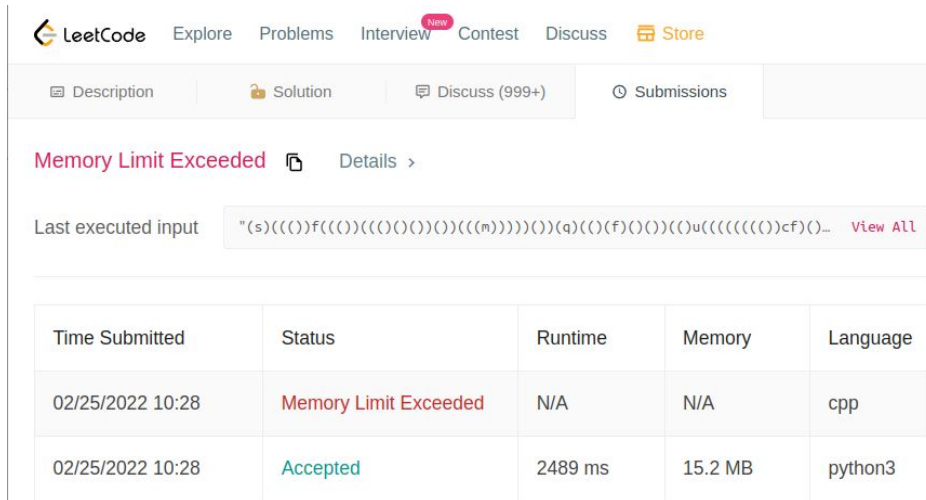
Occurs when the program **uses up the memory assigned** to it in the call stack

In the case of recursion we will mainly deal with two types:

- **Memory Limit Exceeded**
- **Maximum Recursion Depth Exceeded.**

Memory Limit Exceeded

is a type of [stack overflow error](#) that occurs when the parameters of our function [take too much space](#).



The screenshot shows the LeetCode interface for a problem. The top navigation bar includes 'LeetCode', 'Explore', 'Problems', 'Interview', 'Contest', 'Discuss', and 'Store'. Below this, there are tabs for 'Description', 'Solution', 'Discuss (999+)', and 'Submissions'. The 'Submissions' tab is active, showing a submission with the status 'Memory Limit Exceeded'. Below the status, there is a 'Last executed input' field containing a complex string of parentheses and a 'View All' link. At the bottom, there is a table with submission details.

Time Submitted	Status	Runtime	Memory	Language
02/25/2022 10:28	Memory Limit Exceeded	N/A	N/A	cpp
02/25/2022 10:28	Accepted	2489 ms	15.2 MB	python3

Maximum Recursion Depth Exceeded

is a type of [stack overflow error](#) that occurs when the [number of recursive calls](#) being stored in the call stack is [greater than allowed](#).

Different programming languages have different maximum recursion Depth sizes:

- Python => 1000
- Javascript => 10,000
- Java => around 10,000
- C++ => No Limit

```
>>> def count(number):
...     if not number:
...         return number
...     return 1 + count(number - 1)
...
>>> print(count(900))
900
>>> print(count(1000))
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
  File "<stdin>", line 4, in count
  File "<stdin>", line 4, in count
  File "<stdin>", line 4, in count
    [Previous line repeated 996 more times]
RecursionError: maximum recursion depth exceeded
>>> □
```

Python recursion template

```
import sys, threading

input = lambda: sys.stdin.readline().strip()

def main():
    # write your code here

if __name__ == '__main__':

    sys.setrecursionlimit(1 << 30)
    threading.stack_size(1 << 27)

    main_thread = threading.Thread(target=main)
    main_thread.start()
    main_thread.join()
```

A function with missing base case

```
def fib(n: int) -> int:  
    if n == 1:  
        return 1
```

What happens when $n == 0$?

```
    return fib(n-1) +  
    fib(n-2)
```

Wrong recurrence relation

```
def numOfPeopleInQueue(n: int) -> int:  
    if n == 1:  
        return 1  
  
    prev = numOfPeopleInQueue(n + 1) + 1  
    return prev
```

Using list as state

```
def divideInBuckets(i, picked: List) -> int:  
    if i >= len(arr):  
        return 0
```

```
    notPick = divideInBuckets(i+1, picked)
```

```
    picked.append(arr[i])
```

```
    pick = divideInBuckets(i+1, picked)
```

```
    return pick + notPick
```

Passed by
reference

Practice Questions

[Reverse String](#)

[Count Good Numbers](#)

[Pascal's Triangle II](#)

[Merge Two Sorted List](#)

[Decode String](#)

[Predict the Winner](#)

[Power of three](#)

[Power of four](#)

Resources

- [Leetcode Recursion I Card](#)
- [Leetcode Recursion II Card](#)

Quote

“In order to understand recursion, one must first understand recursion.”

Unknown